File

Name: P2-K01-requested-edits
Description: KISSsoft example
Changed by: AAA on: 30.10.2026 at: 19:32:38

Warning: Calculation not consistent.
Results are probably not valid!

1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		0.000.0	
Gear 2:		0.000.0	
		------- Gear 1	---------- Gear 2 -------
Power (kW)	[P]	75.000	
Speed (1/min)	[n]	440.8	145.0
Number of load cycles (in mio.)	[NL]	0.000	0.000
Torque (Nm)	[T]	1624.8	4939.3
Application factor	[KA]	1.25	
Required service life (h)	[H]	0.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Counterclockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1	---------- Gear 2 -------
Center distance (mm)	[a]	303.000	
Center distance tolerance		ISO 286:2010 Measure js6	
Normal module (mm)	[mn]	6.0000	
Transverse module (mm)	[mt]	6.0000	
Normal diametral pitch (1/in)	[Pnd]	4.23333	
Transverse diametral pitch (1/in)	[Ptd]	4.23333	
Normal pressure angle (°)	[αn]	20.0000	
Helix angle at reference circle (°)	[β]	0.0000	
Number of teeth	[z]	23	68
Facewidth (mm)	[b]	28.30	28.30
Hand of gear		Spur gear	
Accuracy grade	[Q-ISO 1328:2013]	A6	A6
Inner diameter (mm)	[di]	0.00	0.00
Shaft diameter (mm)	[ds]	77.58	377.50
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1	---------- Gear 2 -------

3 Materials

Gear 1
18CrNiMo7-6, , , ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

18CrNiMo7-6, , , ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

		------- Gear 1	---------- Gear 2 -------
Surface hardness			
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	425.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	850.00	850.00

3.1 Gear roughness

		------- Gear 1	---------- Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	0.60
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	4.80
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type		
Type of oil	Klüberoil GEM 1-220 N with details about wear coefficient kw	
Lubricant base		
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	220.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	19.00
Specific density at 15°C (kg/dm³)	[ρ]	0.890
Oil temperature (°C)	[TS]	80.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.000
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	71.562
(mm)	[a0.e/i]	70.360 /70.292

Reference profile Gear 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.000
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000

Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	202.509
(mm)	[a0.e/i]	202.330 /202.248

4.1.1 Information on final machining

		------- Gear 1 ----------	Gear 2 -------
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius reference profile	[ρfP*]	0.380	0.380
Addendum reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:			
Tip relief by running in (µm)	[Ca L/R]	0.0 / 0.0	0.0 / 0.0

4.2 Basic data

Overall transmission ratio	[itot]	0.000
Gear ratio	[u]	2.957
Transverse module (mm)	[mt]	6.000
Transverse pressure angle (°)	[αt]	20.000
Working pressure angle (°)	[αwt]	32.150
(°)	[αwt.e/i]	0.000 / 0.000
Working pressure angle at normal section (°)	[αwn]	0.000
Base helix angle (°)	[βb]	0.000
Helix angle at operating pitch circle (°)	[βw]	0.000
Reference center distance (mm)	[ad]	273.000
Sum of profile shift coefficients	[Σxi]	0.0000

		------- Gear 1 ----------	Gear 2 -------
Profile shift coefficient	[x]	0.4270	-0.2485
Generating profile shift coefficient	[xE.e/i]	0.2267/ 0.2153	-0.2783/-0.2920
Virtual gear no. of teeth	[zn]	0.000	0.000
Involute length (mm)	[l_dFa-l_dFf]	0.000	0.000

4.3 Diameters and their allowances

		------- Gear 1 ----------	Gear 2 -------
Reference diameter (mm)	[d]	138.000	408.000
Base diameter (mm)	[db]	129.678	383.395
Tip alteration (mm)	[k*mn]	0.000	0.000
Tip diameter (mm)	[da]	164.982	465.018
(mm)	[da.e/i]	164.982 /164.882	465.018 /464.918
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.100	0.000 / -0.100
Tip form diameter (mm)	[dFa]	0.000	0.000
(mm)	[dFa.e/i]	0.000 / 0.000	0.000 / 0.000
Active tip diameter (mm)	[dNa]	0.000	0.000
(mm)	[dNa.e/i]	0.000 / 0.000	0.000 / 0.000
V-Circle diameter (mm)	[dv]	143.124	405.018
(mm)	[dv.e/i]	140.721 /140.584	404.661 /404.496

Operating pitch diameter (mm)	[dw]	0.000	0.000
(mm)	[dw.e/i]	153.173 /153.157	452.859 /452.811
Root diameter (mm)	[df]	137.982	438.018
(mm)	[df.e/i]	137.721 /137.584	437.661 /437.496
Active root diameter (mm)	[dNf]	0.000	0.000
(mm)	[dNf.e/i]	0.000 / 0.000	0.000 / 0.000
(Indication 0.0 is for dNf.e/i in case of contact interference)			
Root form diameter (mm)	[dFf]	0.000	0.000
(mm)	[dFf.e/i]	0.000 / 0.000	0.000 / 0.000

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.000	0.000
Effective tip clearance (mm)	[c.e/i]	0.000 / 0.000	0.000 / 0.000
Addendum, $m_n(h_{aP}*+x+k)$ (mm)	[ha]	13.491	28.509
(mm)	[ha.e/i]	13.491 /13.441	28.509 /28.459
Dedendum, $m_n(h_{fP}*-x)$ (mm)	[hf]	0.009	-15.009
(mm)	[hf.e/i]	0.140 / 0.208	-14.830 /-14.748
Tooth height (mm)	[h]	0.000	0.000

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	0.000 / 0.000	0.000 / 0.000
Roll angle to dNf (°)	[ξNf.e/i]	0.000 / 0.000	0.000 / 0.000
Roll angle at dFf (°)	[ξFf.e/i]	0.000 / 0.000	0.000 / 0.000

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.8816	1.3899
Normal tooth thickness at tip circle (mm)	[san]	0.000	0.000
(mm)	[san.e/i]	0.000 / 0.000	0.000 / 0.000
Normal tooth thickness at tip form circle (mm)	[sFan]	0.000	0.000
(mm)	[sFan.e/i]	0.000 / 0.000	0.000 / 0.000
Normal space width at root circle (mm)	[efn]	0.000	0.000
(mm)	[efn.e/i]	0.000 / 0.000	0.000 / 0.000
Pitch on reference circle (mm)	[pt]	0.000	0.000
Base pitch (mm)	[pbt]	0.000	0.000
Transverse pitch on contact-path (mm)	[pet]	0.000	0.000

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	0.000	0.000
Specific sliding at the tip	[ζa]	0.000	0.000
Specific sliding at the root	[ζf]	0.000	0.000
Mean specific sliding	[ζm]	0.000	

Sliding factor on tip	[Kga]	0.000	0.000
Sliding factor on root	[Kgf]	0.000	0.000

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	44.000
Transverse contact ratio	[εα]	0.000
	[εα.e/m/i]	0.000 / 0.000/ 0.000
Overlap ratio	[εβ]	0.000
Total contact ratio	[εγ]	0.000
	[εγ.e/m/i]	0.000 / 0.000/ 0.000
Length of path of contact (mm)	[gα]	0.000
(mm)	[gα.e/m/i]	0.000 / 0.000/ 0.000
Approach path of contact (mm)	[gf]	0.000
(mm)	[gf.e/m/i]	0.000 / 0.000/ 0.000
Recess path of contact (mm)	[ga]	0.000
(mm)	[ga.e/m/i]	0.000 / 0.000/ 0.000

		Gear 1	Gear 2
Addendum contact ratio	[ε]	0.000	0.000
	[ε.e/i]	0.000 / 0.000	0.000 / 0.000
Length T1-A and T2-A (mm)	[T1A,T2A]	0.000	0.000
(mm)	[.e/i]	0.000 / 0.000	0.000 / 0.000
Length T1-B and T2-B (mm)	[T1B,T2B]	0.000	0.000
(mm)	[.e/i]	0.000 / 0.000	0.000 / 0.000
Length T1-C and T2-C (mm)	[T1C,T2C]	0.000	0.000
(mm)	[.e/i]	0.000 / 0.000	0.000 / 0.000
Length T1-D and T2-D (mm)	[T1D,T2D]	0.000	0.000
(mm)	[.e/i]	0.000 / 0.000	0.000 / 0.000
Length T1-E and T2-E (mm)	[T1E,T2E]	0.000	0.000
(mm)	[.e/i]	0.000 / 0.000	0.000 / 0.000
Length T1-T2 (mm)	[T1T2]	0.000	
(mm)	[.e/i]	0.000 / 0.000	
Diameter of single contact point B (mm)	[d-B]	0.000	0.000
(mm)	[d-B.e/i]	0.000 / 0.000	0.000 / 0.000
Diameter of single contact point D (mm)	[d-D]	0.000	0.000
(mm)	[d-D.e/i]	0.000 / 0.000	0.000 / 0.000

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	0.00	
Axial force (N)	[Fa]	0.00	0.00
Radial force (N)	[Fr]	0.00	0.00
Normal force (N)	[Fnorm]	0.00	0.00
Nominal circumferential force per mm (N/mm)	[w]	0.00	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]		Division by zero: ZR[0].d/ZPP[0].dw
Axial force (N)	[Faw]	0.00	0.00
Radial force (N)	[Frw]		Division by zero: ZR[0].d/ZPP[0].dw

Circumferential speed reference circle (m/s)	[v]	0.00	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	0.00	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.0	
Running-in value (µm)	[yf]	0.0	
Correction factor	[CM]	0.000	
Gear blank factor	[CR, bs/b, sr/mn]	0.000	(0.000,0.000)
Basic rack factor	[CB]	0.000	
Material coefficient	[E/Est]	0.000	
Single stiffness (N/mm/µm)	[c']	0.000	
Contact stiffness (N/mm/µm)	[cγα]	0.000	
Contact stiffness (N/mm/µm)	[cγβ]	0.000	
The formula for c' and c_γ when w K_A < 25 N/mm is extremely inaccurate!			
c' and c_γ are calculated with w K_A = 25 N/mm.			
Reduced mass (kg/mm)	[mRed]	0.00000	
Resonance speed (min-1)	[nE1]	0	
Resonance ratio (-)	[N]	0.000	
Subcritical range			

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.0	
Bearing distance l of pinion shaft (mm)	[l]	88.000	
Distance s of pinion shaft (mm)	[s]	8.800	
Outside diameter of pinion shaft (mm)	[dsh]	44.000	
Load in accordance with Figure 13, ISO 6336-1:2006	[-]		4
0:a), 1:b), 2:c), 3:d), 4:e)			
With stiffening			
Effective flank line deviation (µm)	[Fβy]	0.00	
from deformation of shaft (µm)	[fsh·B1]	0.00	
fsh (µm)	[fsh]	2.98	
B1	[B1]	0.00	
fHβ5 (µm)	[fHβ5]	0.00	
Tooth without flank line modification			
Position of contact pattern:	favorable		
from production tolerances (µm)	[fmα*B2]	14.14	
B_2=0.00			
Tooth trace deviation, theoretical (µm)	[Fβx]	0.00	
Running-in value (µm)	[yβ]	0.00	

5.4 K factors

Dynamic factor	[Kv]	1.025
Face load factors		
- Flank	[KHβ]	1.054
- Tooth root	[KFβ]	0.000
- Scuffing	[KBβ]	0.000
Transverse load factors		
- Flank	[KHα]	1.000
- Tooth root	[KFα]	0.000
- Scuffing	[KBα]	0.000
Application factor	[KA]	1.250
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: C

		------- Gear 1	---------- Gear 2 -------
Calculated with profile shift coefficient	[x]	0.0000	0.0000
Tooth form factor	[YF]	0.000	0.000
Stress correction factor	[YS]	0.000	0.000
Load application angle (°)	[αFen]	0.000	0.000
Load distribution influence factor	[fε]	0.000	
Load application diameter (mm)	[d_{en}]	0.000	0.000
Bending moment arm (mm)	[hFe]	5.605	6.790
(-)	[hFe*]	0.934	1.132
Tooth root thickness at critical cross section (mm)	[sFn]	0.000	0.000
(-)	[sFn*]	0.000	0.000
Root fillet radius at critical cross section (mm)	[ρF]	0.000	0.000
(-)	[ρF*]	0.000	0.000
Diameter at critical cross section (mm)	[d_{sFn}]	0.000	0.000
Tangent at critical cross section (°)	[α_{sFn}]	0.000	0.000
Notch parameter	[q_s]	Division by zero: ZPP[0].Fuss.sFn/ZPP[0].Fuss.roF	Divisio
Contact ratio factor	[Yε]	0.000	
Helix angle factor	[Yβ]	0.000	
Deep tooth factor	[YDT]	0.000	
Gear rim factor	[YB]	1.00	1.00
Effective facewidth (mm)	[beff]	0.00	0.00
Nominal stress at tooth root (N/mm²)	[σF0]	0.00	0.00
Tooth root stress (N/mm²)	[σF]	0.00	0.00
Permissible bending stress at root of Test-gear			
Notch sensitivity factor	[YδrelT]	0.997	0.995
Surface factor	[YRrelT]	0.957	0.957
Size factor, tooth root	[YX]	0.990	0.990
CHD factor, DNV-CG-0036	[YC]	0.000	0.000
Life factor	[YNT]	0.000	0.000
$Y_C \cdot Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$		0.000	0.000
Alternating bending factor, mean stress influence coefficient			
	[YM]	1.000	1.000
Stress correction factor	[YST]	0.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)	[σFE]	0.00	0.00
Permissible tooth root stress σFG/SFmin (N/mm²)	[σFP]	Division by zero: ZPP[0].Fuss.sigFP/ZS.SSi.Fuss	Division b
Limit strength tooth root (N/mm²)	[σFG]	0.00	0.00

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	0.00	0.00
Safety for tooth root stress	[SF=σFG/σF]	0.00	0.00
Transmittable power (kW)	[kWRating]	0.00	0.00

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	0.000	
Elasticity coefficient (√N/mm)	[ZE]	0.000	
Contact ratio factor	[Zε]	0.000	
ε_α is <1. This case is not described in the standard. Z_ε will be set to 1.0.			
Helix angle factor	[Zβ]	0.000	
Effective facewidth (mm)	[beff]	0.00	
Nominal contact stress (N/mm²)	[σH0]	0.00	
Contact stress at operating pitch circle (N/mm²)	[σHw]	0.00	

Single tooth contact factor	[ZB,ZD]	0.00	0.00
Contact stress (N/mm²)	[σHB, σHD]	0.00	0.00
Lubricant coefficient for N_L	[ZL]	1.020	1.020
Speed factor at N_L	[ZV]	0.974	0.974
Roughness factor for N_L	[ZR]	0.980	0.980
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.000	0.000
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.000	0.000

Limited pitting is permissible: No

Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	Division by zero:	
ZPP[0].Flanke.sigHP/ZS.SSi.Flanke	Division by zero: ZPP[1].Flanke.sigHP/ZS.SSi.Flanke		
Pitting stress limit (N/mm²)	[σHG]	0.00	0.00

7.1 Safety factors

		------- Gear 1 ----------	Gear 2 -------
Required safety	[SHmin]	0.00	0.00
Safety factor for contact stress on operating pitch circle	[SHw]	0.00	0.00
Safety against pressure, σHG/σHBD Single contact	[SHBD]	0.00	0.00
Safety regarding transmittable torque	[SHBD²]	0.00	0.00
Transmittable power (kW)	[kWRating]	0.00	0.00

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018

Lubricant load according to FVA Info sheet 54/7 10, Klüberoil GEM 1-220 N with details about wear coefficient kw

Reference data FZG-C Test:

Torque (Nm)	[T1Ref]	0.000
Line load at contact point A (N/mm)	[FbbRef,A]	0.000
Oil temperature (°C)	[θOilRef]	0.000
Tooth mass temperature (°C)	[θMRef]	0.000
Contact temperature (°C)	[θBRef,A]	0.000
Lubrication gap thickness (µm)	[hRef,A]	0.000
Specific film thickness in test	[λGFT]	0.000

		------- Gear 1 ----------	Gear 2 -------
Calculation of permissible specific film thickness			
Material coefficient	[WW]	0.000	
Permissible specific film thickness	[λGFP]	0.000	

Safety against micropitting:

Calculation has not been carried out, contact analysis under load is required.

9 Tooth flank fracture

Failed to calculate the tooth flank fracture. Please check ISO6336-4_res.TMP file for more information.

10 Scuffing load capacity

		------- Gear 1 ----------	Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	0.000	0.000
Relevant tip relief (µm)	[Ca]	0.00	0.00
Optimal tip relief (µm)	[Ceff]	0.00	
C_a taken as optimal in the calculation. 0=no, 1=yes		0	0

---------- Pair ----------

Lubrication coefficient for lubrication type	[XS]		0.000
Scuffing test, load stage	[FZGtest]	,14	
Relative structural factor, scuffing	[XWrelT]		1.000
Effective facewidth (mm)	[beff]		0.000
Applicable circumferential force/facewidth (N/mm)	[wBt]		0.000
Angle factor	[Xαβ]		0.000

ε_1 =0.000, ε_2 =0.000

10.1 Flash temperature-criteria

		---------- Pair ----------	
Tooth mass temperature (°C)	[θMB]	0.00	
$\theta_{MB} = \theta_{oil} + X_S \cdot 0.47 \cdot \theta_{flamax}$			
Scuffing temperature (°C)	[θS]	0.00	
Γ coordinates (point of highest temperature)	[Γ]	0.000	
Maximum flash temperature (Δ°C)	[θflamax]	0.00	
Maximum contact temperature (°C)	[θBmax]	0.00	
Geometry factor	[XB]	0.000	
Load sharing factor	[XΓ]	0.000	
Dynamic viscosity (mPa*s)	[ηM]	0.00	(80.0 °C)
Coefficient of friction	[μ_m]	0.000	
Required safety	[SBmin]	0.000	
Margin of safety for scuffing, flash temperature	[SB]	0.000	

10.2 Integral temperature-criteria

		---------- Pair ----------
Tooth mass temperature (°C)	[θMC]	0.00
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$		
Integral scuffing temperature (°C)	[θSint]	0.00
Contact ratio factor	[Xε]	0.000
Mean coefficient of friction	[μ_m]	0.000
Geometry factor	[XBE]	0.000
Meshing factor	[XQ]	0.000
Tip relief factor	[XCa]	0.000
Mean flash temperature (Δ°C)	[θflaint]	0.00
Integral tooth flank temperature (°C)	[θint]	0.00
Required safety	[SSmin]	0.000
Safety factor for scuffing (intg.-temp.)	[SSint]	0.000
Safety factor for transmitted moment (int.-T.)	[SSL]	0.000

11 Measurements for tooth thickness

11.1 Tooth thickness tolerances

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.095 / -0.145	-0.130 / -0.190

11.2 Base tangent lengths

		------- Gear 1 ----------	Gear 2 -------
Number of teeth spanned	[k]	4.000	8.000
Base tangent length (no backlash) (mm)	[Wk]	0.000	0.000
Base tangent length with allowance (mm)	[Wk.e/i]	0.000 / 0.000	0.000 / 0.000
(mm)	[ΔWk.e/i]	0.000 / 0.000	0.000 / 0.000
Diameter of measuring circle (mm)	[dMWk.m]	0.000	0.000
Point of contact in profile-modified area (0=no, 1=yes)		1	1

The measurement is inaccurate!

11.3 Measurement over balls and pins

		------- Gear 1	---------- Gear 2 -------
Theoretical diameter of ball/pin (mm)	[DM]	0.000	0.000
Effective diameter of ball/pin (mm)	[DMeff]	11.000	10.000
Radial single-ball measurement, no backlash (mm)	[MrK]	0.000	0.000
Radial single-ball measurement (mm)	[MrK.e/i]	0.000 / 0.000	0.000 / 0.000
Point of contact in profile-modified area (0=no, 1=yes)		1	1
The measurement is inaccurate!			
Diameter of measuring circle (mm)	[dMMr.m]	0.000	0.000
Diametral measurement over two balls, no backlash (mm)	[MdK]	0.000	0.000
Diametral measurement over two balls (mm)	[MdK.e/i]	0.000 / 0.000	0.000 / 0.000
Diametral measurement over pins, no backlash (mm)	[MdR]	0.000	0.000
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	0.000 / 0.000	0.000 / 0.000

11.4 Tooth thickness

		------- Gear 1	---------- Gear 2 -------
Medium tip diameter (mm)	[da.m]	164.932	464.968
Reference chordal height from da.m (mm)	[hac]	0.000	0.000
Tooth thickness at height hac, chord (mm)	[sc]	0.000	0.000
(mm)	[sc.e/i]	0.000 / 0.000	0.000 / 0.000
Tooth thickness on reference circle, arc (mm)	[sn]	0.000	0.000
(mm)	[sn.e/i]	0.000 / 0.000	0.000 / 0.000

The tooth thickness at the reference circle s_n and s_c is output as 0, if the reference circle is located outside the involute area of the tooth flank.
Note: Select "Calculation" -> "Tooth thickness" to calculate the tooth thickness in any diameter.

11.5 Backlash

		---------- Pair ----------
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.020 /-0.020
Radial backlash (mm)	[jrw.e/i]	0.016 /-0.016
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.000 / 0.000
Normal backlash (mm)	[jnw.e/i]	0.000 / 0.000
Center distance allowances (mm)	[Aa.e/i]	0.016 /-0.016
Backlash free center distance (mm)	[aControl.e/i]	303.000 /303.000
Backlash free center distance, allowances (mm)	[jta]	0.000 / 0.000

		------- Gear 1	---------- Gear 2 -------
Tip clearance (mm)	[c0.i(aControl)]	0.016	0.016

12 Gear tolerances

		------- Gear 1	---------- Gear 2 -------
According to ISO 1328-1:2013, ISO 1328-2:1997			
Accuracy grade	[Q]	A6	A6
Single pitch tolerance (µm)	[fpT]	±0.00	±0.00
Base pitch tolerance (µm)	[fpbT]	±0.00	±0.00
Sector pitch tolerance (µm)	[Fpz/8T]	±0.00	±0.00
Cumulative pitch tolerance, total (µm)	[FpT]	0.00	0.00
Adjacent pitch difference tolerance (µm)	[fuT]	0.00	0.00
Profile form tolerance (µm)	[ffαT]	0.00	0.00
Profile slope tolerance (µm)	[fHαT]	±0.00	±0.00
Profile tolerance, total (µm)	[FαT]	0.00	0.00
Helix form tolerance (µm)	[ffβT]	0.00	0.00
Helix slope tolerance (µm)	[fHβT]	±0.00	±0.00

Helix tolerance, total (µm)	[FβT]	0.00	0.00
Runout tolerance (µm)	[FrT]	0.00	0.00
Single flank composite tolerance, total (µm)	[FisT]	0.00	0.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	0.00	0.00
Radial composite tolerance, total (µm)	[Fi'']	0.00	0.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi'']	0.00	0.00

F_{idT} (F_i''), f_{idT} (f_i'') according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020			
Accuracy grade	[R]	R41	R41
Radial composite tolerance, total (µm)	[FidT]	0.00	0.00
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	0.00	0.00

13 Modifications and determination of the tooth form

13.1 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

14 Supplementary data

14.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (g)	[m]	0.00	0.00
Total mass (g)	[Σm]	0.00	
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.01658	0.71892

14.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (MNm/rad)	[cr]	0.000
Torsion when subjected to nominal torque (°)	[δcr]	0.000
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	78.219

14.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.000
Wear sliding coefficient by Niemann	[ζw]	0.000
Loss factor	[HV]	0.000
Gear power loss (kW)	[PVZ]	0.000

15 Service life, damage

Could not be calculated.

16 Remarks

16.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch

circle.

16.2 Calculations and factors

- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
  $K_{H\beta}$ and $K_{F\beta}$ according to Method C
  $f_{m\alpha}$ according to equation 66, f_{sh} according to 59/60, $F_{\beta x}$ according to 54/55/57
  $K_{H\alpha}$, $K_{F\alpha}$ according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 573)
